Automated Network Request Management

Phase 1: Ideation Phase — Business Strategy, Architecture Vision & Stakeholder Alignment

1. Executive Summary & Vision Statement

Modern enterprise IT environments demand agile, reliable, and standardized network operations. Historically, network provisioning—ranging from port allocations, IP assignments, firewall rule modifications, and bandwidth adjustments—has suffered from fragmented communication channels (informal emails, chats, disparate spreadsheets). This project establishes an automated, end-to-end **Automated Network Request Management System** built natively on the **ServiceNow** platform.

> **Project Mission:** Streamline and automate the complete lifecycle of network-related service requests using ServiceNow, eliminating manual friction, enforcing rigorous security approvals, accelerating fulfillment SLA, and providing 100% auditable visibility.

2. Problem Statement & Baseline Operational Challenges

Prior to this automated solution, network service intake across the enterprise exhibited severe operational bottlenecks:

- **Manual Effort & Human Error:** Requesters submit incomplete network requirements via emails or unvalidated forms, leading to endless back-and-forth communication and configuration inaccuracies on physical network infrastructure.
- **Protracted Turnaround Times:** The average cycle time from initial request to technical execution extended across multiple business days due to ad-hoc, manual email approval chasing.
- **Lack of Unified Visibility:** Stakeholders possessed zero real-time status tracking, leading to repeated status-inquiry tickets and escalations.
- **Security & Audit Vulnerabilities:** Ad-hoc approvals failed to comply with strict SOC2/ISO27001 ITIL compliance standards, lacking immutable audit trails.

3. Strategic Business Objectives & Quantifiable Goals

Strategic Objective	Target Outcome	Business Impact
Manual Toil Reduction	Eliminate manual data re-entry across teams	80% reduction in administrative ticket handling
SLA Acceleration	Standard requests fulfilled within 2 business days	70% faster turnaround time from submission to closure
Policy Governance	Automate role-based managerial approvals	100% compliance with corporate network security policies
Auditable Traceability	Centralize transactions on custom platform tables	Zero audit non-conformances with immutable logs
Self-Service UX	Modern ServiceNow Service Portal catalog interface	Enhanced employee satisfaction (CSAT > 92%)

4. Stakeholder Matrix & Impact Assessment

A multi-disciplinary stakeholder group interacts with the Automated Network Request Management lifecycle. The table below details their core responsibilities, operational expectations, and the direct impact delivered by automation:

Stakeholder	Role & Function	Needs & Expectations	Impact of Automation
End Users (Requesters)	Corporate employees requesting network provisioning, IP access, or connectivity.	• Simple, intuitive request portal • Dynamic field visibility • Real-time request tracking	✔■ Rapid self-service submission ✔■ Automated email status notifications ✔■ Transparent delivery estimates
IT Administrators	Platform architects managing ServiceNow configurations, workflows, and ACLs.	• Reliable no-code flow logic • Standardized data structures • Minimal ongoing maintenance	✔■ Reduced administrative ticket load ✔■ Out-of-the-box Flow Designer logic ✔■ Clean upgrade paths
Network Team (Fulfillers)	Network engineers executing infrastructure changes and configurations.	• Complete, validated data intake • Elimination of ad-hoc emails • Standardized tasks	✔■ Automated task & record creation ✔■ Zero missing requirement chasing ✔■ Reclaim hours for complex engineering
Approvers (Managers/Security)	Department managers and network security officers authorizing requests.	• Instant notification of requests • Full business justification context • Single-click approval	✔■ Automated approval routing ✔■ Complete audit history ✔■ Reduced approval bottlenecks

5. Solution Scope & Architecture Vision

To ensure rapid time-to-value and enterprise maintainability, the project scope was structured around ServiceNow native capabilities without requiring fragile custom script includes or legacy business rules:

In-Scope Capabilities:

• **Service Catalog Intake:** Creation of a dedicated 'Network Request' catalog item published under the Service Portal (/sp).

• **Dynamic Client UX:** Implementation of Catalog UI Policies and Variable Sets to dynamically display fields (e.g., reveal Existing ID when Connection Type is 'Existing').

• **Custom Transactional Data Model:** Definition of u_database_tables to store validated network requests with auto-numbering.

• **Relationship & Approvals Integration:** Creation of a custom relationship linking u_database_tables to sysapproval_approver for related list visibility.

• **Flow Designer Orchestration:** No-code automated workflow handling variable extraction, record generation, managerial approval routing, email notifications, and state updates.

• **Security & Governance:** Application of Role-Based Access Control (ACL) rules and comprehensive audit logging.

Future Scope (Roadmap Opportunities):

• Automated switch port and VLAN provisioning via Cisco DNA Center or Ansible Playbooks using ServiceNow Integration Hub Spokes.

• Interactive Performance Analytics dashboards measuring request volumes, fulfillment velocities, and resource allocation trends.

• Expansion of automated request fulfillment across edge devices, firewall policy adjustments, and cloud VPC provisioning.

6. Project Success Metrics & Target KPIs

Metric Category	Baseline (Manual)	Automated Target	Measurement Source
Request Cycle Time	72 - 96 Hours	< 24 - 48 Hours	ServiceNow Task Duration Logs
First-Time Accuracy	64% complete submissions	> 98% validated submissions	Mandatory Field Enforcements
Approval Turnaround	36 Hours (Manual Email Chasing)	< 4 Hours (One-Click Flow)	sysapproval_approver Audit Log
Audit Compliance	Inconsistent spreadsheet logs	100% Platform Traceability	ServiceNow System Activity Logs

Phase 1 Sign-Off: The ideation phase establishes clear business alignment, stakeholder consensus, and architectural boundaries. The initiative proceeds to Phase 2 (Project Design) for data modeling and UI logic definition.